2 - Here we have two assumptions and two formula

If we are assuming that the two end points are not included

Then; $\frac{|b-a|}{2^{n+1}}$

If we are assuming either endpoint of interval is included

Then, $\frac{|b-a|}{2^{n}}$

So, we have;

$$n \geq \frac{\ln\left(\frac{b-a}{\epsilon}\right)}{\ln 2}$$

$$\geq 31.48$$

$$\boxed{n = 32}$$

3. Secant's method

$$x_{n+1} = x_n - \left(\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}\right) f(x_n)$$

$$= 1 - \left(\frac{1-0}{1-2}\right) \times 1$$

$$= 1 + 1$$

$$\boxed{= 2}$$